

EXPLORATORY DATA ANALYSIS (EDA)

Project 2 — EDA Insights Report

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About This Project

This report documents the Exploratory Data Analysis (EDA) performed on the DecodeLabs e-commerce order dataset (1,200 records x 14 columns). The goal was to uncover hidden patterns, trends, and outliers to transform raw numbers into meaningful business insights. Python (pandas) was used in Jupyter Notebook for all analysis.

Dataset Overview

Attribute	Value
File Name	Dataset for Data Analytics.xlsx
Total Records	1,200 rows
Total Columns	14 columns
Date Range	2023-01-01 to 2025-06-30
Tool Used	Python (pandas) in Jupyter Notebook
Analysis Date	May 06, 2026

Key Findings — Final Summary

Metric	Value	Insight
Total Orders	1,200	Full dataset analyzed
Total Revenue	Rs. 12,64,761.96	Strong business performance
Average Order Value	Rs. 1,053.97	Healthy per-order revenue
Most Sold Product	Printer (181 orders)	Highest demand product
Least Sold Product	Phone (156 orders)	Needs marketing push
Most Expensive Product	Laptop (avg Rs.1,110)	Premium category leader
Top Payment Method	Online (258 orders)	Digital payments preferred
Top Referral Source	Instagram (259 orders)	Social media drives sales
Best Sales Month	June 2024 (Rs.68,068)	Peak revenue month
Worst Sales Month	April 2024 (Rs.27,909)	Lowest revenue month
Outliers Found	8 high-value orders	Possible VIP/bulk customers

Cancelled Orders	250 (20.8%)	High — needs investigation
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Product Analysis

Product	Orders Count	Avg Revenue (Rs.)	Rank
Printer	181	1,080.73	#1 Best Seller
Tablet	179	1,042.28	#2
Chair	178	1,098.99	#3
Laptop	173	1,110.56	Most Expensive
Desk	170	985.06	#5
Monitor	163	1,077.62	#6
Phone	156	972.58	Cheapest / Least Sold

Outliers Detected

Using the IQR method ($Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$), the normal price range was found to be Rs.-1,341 to Rs.3,330. Eight orders were found above this range — these are high-value outliers that could represent VIP customers or bulk purchases.

Order ID	Product	Total Price (Rs.)	Status
ORD200107	Printer	3,353.75	High Value
ORD200326	Laptop	3,352.40	High Value
ORD200328	Tablet	3,370.20	High Value
ORD200469	Chair	3,384.90	High Value
ORD200632	Laptop	3,390.80	High Value
ORD200789	Tablet	3,456.40	Highest Value
ORD201065	Printer	3,334.00	High Value
ORD201122	Monitor	3,390.95	High Value

Business Recommendations

#	Finding	Recommendation
1	Cancelled orders are 20.8% (250)	Investigate cancellation reasons; improve delivery & customer service
2	Instagram drives most customers (259)	Increase Instagram marketing budget for better ROI
3	Phone is least sold product	Run promotions or discounts on Phone category
4	June 2024 was best month	Analyze what worked in June and replicate strategy

5	8 high-value outlier orders	Create VIP customer program to retain these buyers
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Conclusion

The EDA revealed that the DecodeLabs e-commerce business generates strong revenue of Rs.12.6 Lakhs from 1,200 orders. Printers are the top-selling product while Instagram is the strongest customer acquisition channel. The most critical business concern is the high cancellation rate of 20.8% which requires immediate attention. Eight high-value outlier orders represent potential VIP customers that should be nurtured with a loyalty program.

PROJECT 2 EDA COMPLETE — All Insights Verified ■